# D3.3 Homeostasis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d3-3.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Image credits are on the last slide of the slides for this subtopic.

## The principle

### D3.3.1 Homeostasis: keeping the internal environment steady

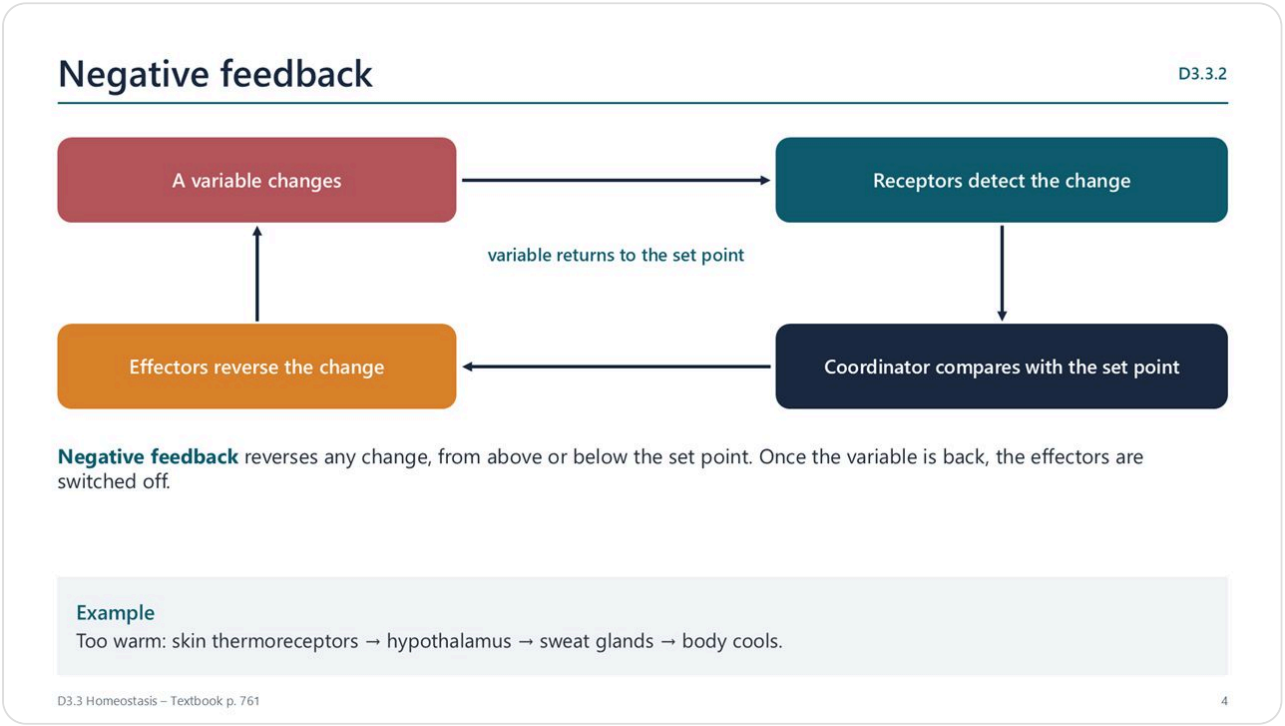
p. 760

**Homeostasis** is maintaining the internal environment within preset limits, despite changes in the external environment. In humans, the controlled variables include **body temperature**, **blood pH**, **blood glucose concentration** and **blood osmotic concentration**.

**Watch out:** homeostasis is about the internal environment. It does not mean moving to somewhere more comfortable.

### D3.3.2 How negative feedback loops work

p. 761




A receptor detects a change, a coordinator compares it with the set point, and an effector acts in the opposite direction. **Negative feedback** returns a variable to the set point from above and from below. Positive feedback would move it further away, which is why homeostasis does not use it.

# Blood glucose

## D3.3.3 How hormones regulate blood glucose

pp. 761–763

Blood glucose is normally about 90 mg per 100 $cm^3$. The brain has no glycogen store, so it depends on a steady supply from the blood.

| | Glucose too high | Glucose too low |
|---|---|---|
| **Detected by** | Beta cells in the islets of Langerhans | Alpha cells in the islets of Langerhans |
| **Hormone** | Insulin | Glucagon |
| **Target cells** | Liver, muscle and other tissues | Liver |
| **Effect** | Glucose uptake; storage as glycogen; more used in respiration | Glycogen broken down; glucose released |

**Watch out:** glucagon is a hormone, and glycogen is the storage product. The pancreas itself detects the change, so the hypothalamus is not involved.

## D3.3.4 The changes behind type 1 and type 2 diabetes

pp. 764–765

| | Type 1 | Type 2 |
|---|---|---|
| **Physiological change** | The immune system destroys beta cells, so little or no insulin is produced | Insulin is produced, but target cells do not respond well to it |
| **Risk factors** | Genes, and possibly exposure to some viruses | Overweight, inactivity, age, family history |
| **Prevention** | None known at present | Healthy diet, regular exercise, healthy weight |
| **Treatment** | Insulin, with regular monitoring of blood glucose | Diet and exercise, sometimes medicine |

In either type, blood glucose stays raised and glucose may appear in urine. Long-term effects can include damage to the kidneys, eyes, nerves and blood vessels.

**Watch out:** do not say that type 2 is simply caused by a high-sugar diet. Such a diet is linked with it, and is one of several risk factors.

# Body temperature

## D3.3.5 Temperature control as negative feedback

p. 766

**Peripheral thermoreceptors** in the skin detect temperature changes. The **hypothalamus** compares these with the set point and signals to effectors, including muscle and adipose tissue.

Hormones are involved too. The hypothalamus secretes a releasing hormone, the **pituitary gland** secretes thyroid-stimulating hormone, and the thyroid gland secretes **thyroxin**. Thyroxin raises oxygen consumption and metabolic rate, which generates heat.

### Thyroxin controls heat production

D3.3.5




| Effector tissue | Role |
|---|---|
| **Skeletal muscle** | Shivering releases heat |
| **White adipose tissue** | Insulates: a long-term response to cold |
| **Brown adipose tissue** | Releases heat fast by uncoupled respiration |

High thyroxin reduces secretion by the hypothalamus and pituitary: negative feedback again.

D3.3 Homeostasis – Textbook p. 766 18

## D3.3.6 How humans regulate body temperature

pp. 766–767

| | Too hot | Too cold |
|---|---|---|
| **Skin arterioles** | Vasodilation: more blood to the skin | Vasoconstriction: less blood to the skin |
| **Sweat glands** | More sweat, which evaporates | Less sweat |
| **Hair** | Muscles relax; hairs lie flat | Muscles contract; hair erection traps air |
| **Muscle** | No shivering | Shivering generates heat |
| **Brown adipose tissue** | Little activity | Uncoupled respiration releases heat |

Sweat cools the skin because evaporation breaks hydrogen bonds, which takes heat energy. **Uncoupled respiration** in brown adipose tissue releases energy as heat rather than storing it in ATP, which is important for newborn babies. Birds and mammals also use behaviour to regulate temperature.

**Watch out:** blood vessels do not move up or down in the skin. Arterioles widen or narrow, which changes how much blood flows through the surface capillaries.

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